STANFORD WHITE IV

stanfordwhiteiv@gmail.com github.com/stanfordwhiteiv

EDUCATION

National Science Foundation Graduate Research Fellow (NSF–GRFP) *Aug 2023 - Present*
University of Mississippi – One of 2,000 selected from 15,000+ applicants nationwide

University of Mississippi
Ph.D. Student in Engineering Science (Exp. 2028) | M.S. in Engineering Science (2025) | GPA: 4.0
B.S. in Mechanical Engineering (2023) | GPA: 4.0 | Summa Cum Laude | Undergraduate Class Marshal

SKILLS

ML Methods: Vision Transformers, Self-Supervised Learning, Foundation Models, Diffusion Models, Physics-Informed ML, Video Prediction, Multi-Objective Optimization, Agentic Workflows
Frameworks & Tools: Python, PyTorch, TensorFlow, Scikit-Learn, Arduino, MATLAB, Docker, Git, ROS, Slurm/HPC
Domain Expertise: Computer Vision, Materials Informatics, Additive Manufacturing, Microstructure Characterization, High-Speed Imaging, Robotics, Blast & Impact Dynamics

RELEVANT EXPERIENCE

Self-Supervised Foundation Model for Polycrystalline Microstructure Representation *Nov 2025 - Present*
University of Mississippi, Oxford, MS
– Pre-trained a ViT-B foundation model with DINO/iBOT++ self-distillation on 217,000+ electron backscatter diffraction (EBSD) scans, using $\mathrm{SO}(3)$ crystal orientations as input with learned invariance to crystal symmetry
– Achieved a median 8.4x label-efficiency improvement over supervised ResNet-50 and a mean 30% accuracy gain on 19 downstream microstructure characterization tasks over DINOv3, MAE, and ResNet-101

Machine Learning Guided Additive Manufacturing *Aug 2023 - Present*
University of Mississippi, Oxford, MS
– Built a 60,000 fps high-speed imaging system and trained a 3D CNN-Transformer for next-frame prediction of electrohydrodynamic jet printing dynamics ($R^2 > 0.90$), replacing CFD simulations with 1000x faster inference
– Applied ML-guided optimization to metal AM and inkjet biofabrication, achieving the first 50/50 duplex steel via laser powder bed fusion, 2x more feasible DED conditions than DOE, and 4% $\rightarrow$ >55% bioink ejection success

Physics-Informed Transformer for Catalytic Pyrolysis of Polymer Nanocomposites *Aug 2024 - Dec 2025*
University of Mississippi & Center for Graphene Research & Innovation, Oxford, MS
– Developed a physics-informed transformer embedding Avrami-Erofeev kinetics to predict char mass evolution in catalytic polymer pyrolysis, achieving $R^2 > 0.98$ on unseen catalysts and extrapolating to untested compositions

Machine Learning for Swarm Robotics (NSF–REU) *May 2022 - June 2023*
West Virginia University & National Science Foundation, Morgantown, WV
– Developed a genetic learning approach to decompose and imitate decentralized swarm behaviors, recovering agent-level policy weights, communication laws, and motion constraints with >90% accuracy

High-Throughput Split-Hopkinson Pressure Bar (SHPB) Data Analysis Software *Jan 2023 - Present*
Personal Project – Targeting Patent Submission
– Built and deployed Python software automating high strain-rate test post-processing, increasing throughput 13x

SELECTED PUBLICATIONS & PRESENTATIONS

White, Stoddard, et al. "Investigating Nozzle-Based Additive Manufacturing Dynamics with High-Speed Imaging and Machine Learning" *Progress in Additive Manufacturing* (IF: 5.2), Sep. 2026.

White, Salvati, et al. "Physics-informed Machine Learning Prediction of Char Mass Evolution in the Catalytic Pyrolysis of Polyetherimide/Graphite Nanocomposites" *Chemical Physics Impact* (IF: 5.3), Dec 2025.

White, Herring, et al. "A Universal Self-Supervised Representation Framework for Polycrystalline Matter" *In Preparation*, Oct. 2026. (Ready to Submit)

White, Rahman, Crockett, et al. "Imitating Swarm Behaviors by Learning Agent-Level Controllers" *2023 Proceedings of the American Control Conference (ACC)*, San Diego, CA, 2023, pp. 3657-3662.

Presentations: MRS Fall 2026 | IMAT 2026 | TMS 2025 | ACC 2023